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Project Title: Comparing Methods for Respiratory Waveform Extraction from Lung MRI

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Identifier	Roles, Tasks, etc.
{AA}	Meetings Lead, Jacobian Researcher and Pipeline Developer and Chief Editor
{BB}	Organisation Lead, Jacobian Researcher, Robustness Pipeline Developer
{CC}	Meetings Communications Lead, ML and ROI Researcher and Agreeability Pipeline Developer and Editor
{DD}	ML Researcher and Usability Pipeline Developer
{EE}	Further Research Developer

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Comparing Methods for Respiratory Waveform Extraction from Lung MRI Images

Abstract: This project compared three methods for extracting respiratory waveforms from lung MRI: Manual Region Of Interest (ROI), Automated Machine Learning (ML), and a Deformation-based method (Jacobian), on usability, robustness, and agreement. Usability was assessed based on runtime, setup effort, reproducibility, and computing demand. Agreement was evaluated on correlation, respiratory rate, cycle count, and detection of abnormal breathing. After the addition of Gaussian blur and Rician noise, robustness was analysed on correlation, root mean squared error and peak timing error. For agreeability, key findings included the inverse correlation between deformation-based and intensity-based (ROI and ML) waveforms, reflecting volume change versus pixel intensity, and respiratory rates that were consistent with physiology across methods (~14 bpm). For usability, the Manual ROI was fastest (37 s/slice) and easiest to integrate, while the Jacobian method was slowest (289 s/slice), and ML had the highest setup work, but showed better scalability for bigger datasets. In robustness, the Jacobian method stayed accurate under ideal conditions but failed beyond moderate noise, while ML showed graceful degradation throughout, with Manual ROI keeping waveform shape but losing temporal precision. No single method was universally superior across all metrics; therefore, choice depends on priorities in scalability, physiology and image quality.

I. Introduction

Magnetic Resonance Imaging (MRI) respiratory waveform extraction methods are widely used for applications such as preprocessing in quantitative lung MRI, where they help reduce motion-induced blurring, and for functional lung assessment. MRI is a non-invasive, non-ionising imaging modality that produces dynamic time-series data. Several extraction approaches exist, including manual region of interest (ROI)-based, machine learning (ML)-based, and deformation-based methods, yet their performance has not been systematically compared under consistent conditions. This motivates an investigation into which methods are most suitable under different constraints and how to ensure accurate, reliable outputs.

All waveform extraction techniques aim to map image sequences into a respiratory waveform by tracking features that vary with breathing, such as motion. These methods differ primarily in feature selection and mask definition. A mask selects regions of interest within an image, mathematically represented as an image-sized function. This study used 2D MRI time-series data comprising 256 frames from two healthy volunteers.

Manual ROI-based extraction methods assume that pixel intensity changes within a selected region correlate with respiration [1]. An ROI is a set of pixels, Ω_{ROI}, typically at maximum lung inflation, and defines the lung region.

This means that this method relies on a human observer to define the spatial boundaries of the lungs.

It is defined as a binary mask, $M_{ROI}(x,y)$, where x and y are pixels in the image:

$$M_{ROI}(x,y) = \begin{cases} 1, & if\ (x,y)\ belongs\ to\ ROI \\ 0, & if\ not \end{cases} \quad (1)$$

The pixel intensity in this region is then taken across all frames and averaged to produce a respiratory waveform.

$$W_{ROI}(t) = \frac{\sum_{(x,y)\in\Omega ROI} I(x,y,t)}{|\Omega_{ROI}|} \quad (2)$$

Equation 2 shows that $I(x,y,t)$is the image intensity at pixel (x, y) in frame t, and $|\ \Omega_{ROI}\ |$ is the number of pixels in the ROI [3]. This approach assumes that the same set of pixels remains within the lung region across frames, which works best when motion is small or when the ROI is combined with motion correction [4].

ML-based extraction methods use automated, data-driven approaches to identify and segment regions of interest without manual input [5]. In this study, a convolutional neural network (CNN) was used to learn spatial features and generate masks, where lung pixels are labelled as 1 and non-lung pixels as 0, producing a binary segmentation. These masks are then averaged across frames to compute intensity changes and generate a respiratory waveform [6]. A pre-trained model was used, avoiding the time and computational cost of training from scratch [5].

Deformation-based extraction methods are uncommonly used in lung MRI and are more typical in Computed Tomography (CT). However, CT is invasive and requires breath-holding [7], making it unsuitable for repeated or patient-friendly use, especially in paediatric groups. Due to limited research on applying deformation-based methods to MRI, AA developed a Python-based pipeline to generate waveforms using this approach.

Deformable image registration measures how local regions of the image expand and contract over time [8]. First, a reference frame is segmented with a mask to which each frame is registered. This estimates a displacement field:

$$\Phi(x,y) = \big(x + u(x,y), y + v(x,y)\big) \quad (3)$$

Where u (xy) is the displacement in the x-direction and v (x,y) is the displacement in the y-direction [9].

The field contains a motion vector for every pixel and represents how lung tissue moves relative to the reference frame.

The Jacobian determinant is then estimated for this deformation field and measures local area change caused by breathing:

$$J(x, y) = \det \begin{vmatrix} \frac{\partial x'}{\partial x} & \frac{\partial x'}{\partial y} \\ \frac{\partial y'}{\partial x} & \frac{\partial y'}{\partial y} \end{vmatrix} \quad (4)$$

Equation 4 can be interpreted as a J > 1 indicates local expansion (inhalation), J < 1 indicates local contraction (exhalation), and J = 1 indicates no change [10]. To obtain a global respiratory waveform, the Jacobian values are averaged across the mask.

II. Methodology

CC extracted manual ROI and automated (ML) mask waveforms for the healthy volunteers (HV), HV_002 (slices 19, 22, 25) and HV_003 (slices 18, 21), which were merged with Jacobian outputs from AA's pipeline to create the analysis dataset. BB developed the robustness pipeline, and DD developed the usability pipeline, also using this data.

Agreeability (CC):

CC developed a MATLAB pipeline to quantify agreement between the three methods (manual–automated, manual–deformation, automated–deformation), using approaches commonly applied in respiratory waveform analysis [11][12][13][14][15]. For each pair of signals, denoted $x(t)$ and $y(t)$ of length n with sampling interval T_A, the following metrics were computed:

- **Pearson correlation** r, which measures the linear relationship between waveforms [11].

- **Mean Absolute Error:**

  $$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} | x_i - y_i | \ (5)$$

  for the average point-wise intensity difference [11] (excluding Jacobian-intensity pairs due to scale differences). All other metrics are scale invariant.

- **Phase agreement** is the proportion of time points where the direction of change (inspiration/expiration) matches:

  $$\frac{1}{n-1}\sum_{i=1}^{n-1} \mathbb{1}[\text{sign}(\Delta x_i) = \text{sign}(\Delta y_i)] \ (6)$$

- **Respiratory rate** (RR, breaths/min):

  $$\text{RR} = 60/\text{mean}(\Delta t_{\text{EE}}) \ (7),$$

  where Δt_{EE} are intervals between consecutive End Expiration (EE) peaks [11].

- **Cycle count** is the number of detected EE peaks.

EE was selected as the reference for both RR and cycle counts because lung conspicuity is optimised at end expiration [16]. MATLAB's 'findpeaks' function identified peaks (EE), using a minimum peak distance of 3 seconds. This threshold was chosen based on the upper-bound RR of 20 bpm [17], and to minimise higher frequency cardiac noise [15][18].

Agreeability - Abnormal Breathing:

Abnormal intervals were independently identified for each method using an RR outlier approach, as RR is a key indicator of lung function [14]. Instantaneous RR was derived from end-expiration peaks, flagging cycles that deviated from the median by more than twice the median absolute deviation (MAD) as abnormal. MAD was chosen as it provides a robust approach to outlier detection for cycle-based time series data [19]. The time indices of these flagged cycles were recorded and compared across all methods to evaluate whether they agreed on what constitutes abnormal breathing.

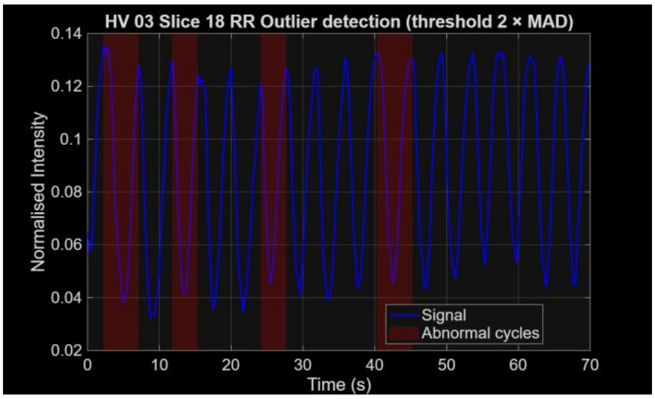

Fig. 1 *shows an example of abnormal breathing detection.*

Usability (DD):

DD developed a quantitative usability framework to assess three methods, applying usability engineering principles rather than relying on qualitative descriptions [20].
For each method, an overall usability utility score U_i was defined as

$$U_i = w_T S_T + w_S S_S + w_R S_R + w_C S_C \quad (8)$$

where S_T, S_S, S_R, and S_C represent the scores for time efficiency, setup burden, reproducibility, and computing-resource demand, respectively. The weights were chosen as $w_T = 0.3$, $w_S = 0.2$, $w_R = 0.3$, and $w_C = 0.2$, so that runtime and reproducibility contributed most to the final score.
Time efficiency was calculated relative to a reference runtime of $T_{ref} = 240$s:

$$S_T = \max\left(0, 1 - \frac{T_i}{T_{ref}}\right) \quad (9)$$

So, methods with longer runtimes received lower scores. Setup burden was calculated from both setup time $(t_{setup})$and the number of manual steps (n_{setup}), using reference values of $t_{ref} = 600$s and $n_{ref} = 10$:

$$S_S = 1 - 0.5\frac{t_{setup}}{t_{ref}} - 0.5\frac{n_{setup}}{n_{ref}} \quad (10)$$

Reproducibility was quantified using the intraclass correlation coefficient (ICC)[21]

$$S_R = \text{ICC} \quad (11)$$

with

$$\text{ICC} = \frac{\sigma^2_{\text{subject}}}{\sigma^2_{\text{subject}} + \sigma^2_{\text{operator}} + \sigma^2_{\text{error}}} \quad (12)$$

Where a higher ICC indicates stronger agreement between operators CC and DD across three methods. Finally, the computing-resource score was calculated by normalising hardware burden (h) and software burden (s) against a reference value of 6:

$$S_C = 1 - \left[0.5\left(\frac{h}{6}\right) + 0.5\left(\frac{s}{6}\right)\right] \quad (13)$$

So, methods requiring greater computational resources received lower scores. Reference times were chosen from the maximum across all methods. U_i , S_T , S_S , and S_C were developed following the composite indicator methodology.

Robustness (BB):

Robustness was defined as the ability of each method to tolerate noise in the input images while preserving the fidelity of the resulting waveform. To evaluate this, controlled levels of noise were applied to the original MRI slices, generating a series of degraded image sets. Respiratory waveforms were then computed from each noise level using all three methods.

Waveform comparisons were evaluated using three complementary metrics. The Pearson correlation coefficient (PCC) assessed similarity in waveform shape, where x_t represents the reference waveform and y_t the waveform obtained from noisy data and is defined as [22]:

$$PCC = \frac{\Sigma_t(x_t - \bar{x})(y_t - \bar{y})}{\sqrt{\Sigma_t(x_t - \bar{x})^2}\sqrt{\Sigma_t(y_t - \bar{y})^2}} \quad (14)$$

PCC ranges from -1 to 1, with 1 indicating perfect agreement, 0 no correlation, and -1 inverse behaviour.

The root mean squared error (RMSE) quantified absolute differences in signal amplitude, where N is the number of frames [23]:

$$RMSE = \sqrt{\frac{1}{N}\Sigma_{t=1}^{N}(x_t - y_t)^2} \quad (15)$$

RMSE is non-negative, with lower values indicating closer agreement between noisy and reference waveforms.

Peak timing error (PTE) evaluated the temporal alignment of respiratory cycles, where $p_k^{(x)}$ and $p_k^{(y)}$ denote the indices of corresponding peaks in the reference and noisy waveforms, and K is the number of matched peaks [24]:

$$PTE = \frac{1}{K}\Sigma_{k=1}^{K}\left|p_k^{(x)} - p_k^{(y)}\right| \quad (16)$$

PTE is measured in frames, with smaller values reflecting more accurate detection of respiratory cycle timing.

Two types of degradation, Rician noise (with values between 0 and 0.2) and Gaussian blur (with σ values between 0 and 2), were investigated to provide a broad and realistic representation of image quality deterioration in MRI. This is because noise in MRI images is commonly modelled using a Rician distribution, and a Gaussian blur was used to simulate loss of both spatial resolution and motion-related artefacts [25][26]. Degradation levels were selected to span from ideal conditions to severe but interpretable image degradation, with the upper bound 2 for Rician chosen such that images approached the low signal-to-noise ratio regime (SNR < 2), where anatomical structure becomes difficult to distinguish, and intensity distortion becomes significant [26][27].

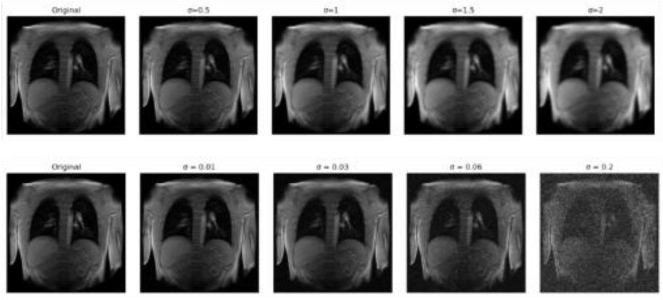

***Fig 2**. shows increasing a)blur b)noise*

III. Results and Analysis

Agreeability (CC):

Patient Dependent Performance

As shown in Fig. 2, manual-automated (man-auto) correlation varied between patients: HV_002 showed moderate agreement (mean of 0.56 ± 0.11 slice variability) while HV_003 was excellent (0.93 ± 0.03). In HV_002, automated detected fewer cycles (14.7 ± 1.5 vs. 17.7 ± 1.2) and a larger RR difference (2.70 bpm), whereas HV_003 remained consistent. These findings align with patient variability found in MRI-derived RR, emphasising that testing on diverse populations is crucial for generalisability [28]. Deformation-intensity signals also showed patient variability, particularly in correlation, as detailed next.

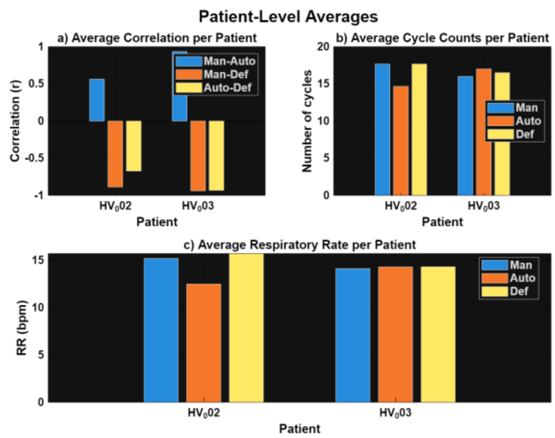

***Fig 3** highlights variability across patient averages for a) correlation, b) cycle counts and c) RR*

Inverse Correlation

The deformation-manual signals exhibited a strong negative correlation and low phase agreement (**Fig. 3**), showing that they moved in opposite directions across both patients. Interestingly, deformation-automated correlation showed patient dependence with HV_003 exhibiting a strong negative r with low phase agreement, whereas HV_002 showed a moderate negative r and phase agreement.

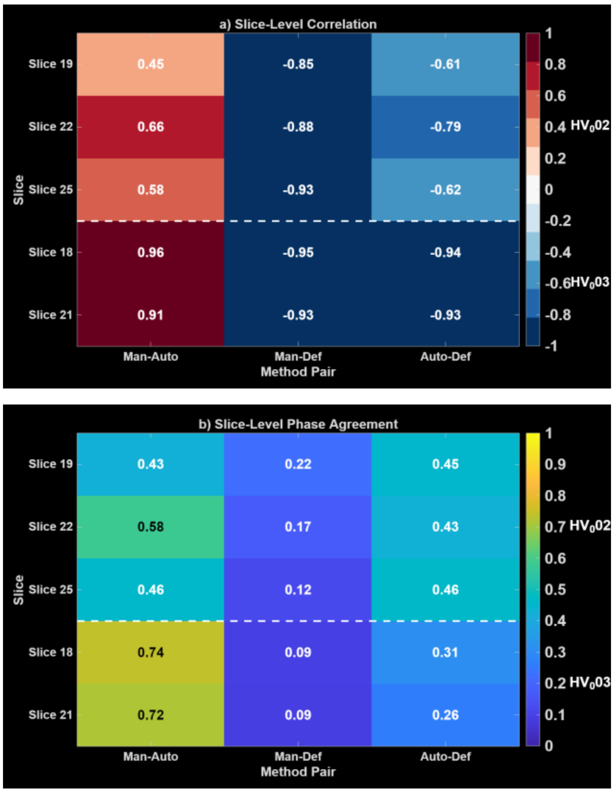

Fig 4 shows a) pairwise correlation, b) phase agreement heatmaps.

This inverse relationship is explained physiologically, as the Jacobian determinant indicates local volume changes (J > 1 during inhalation, J < 1 during exhalation) while intensity-based methods monitor mean pixel intensity, where inhalation darkens the lung parenchyma, resulting in decreased intensity [29][30][31][32]. Thus, the peaks and troughs of these two signal types are inverted. Lastly, the MAE between the intensity-based methods was minimal.

Respiratory Rate

Despite patient variability, RR estimates across methods (median 14.2 bpm) aligned with normal adult breathing rates [33]. Furthermore, inter-method agreement (mean absolute difference 0.09–3.77 bpm) compared favourably to ECG-derived RR MAE (3.36–5.79 bpm), suggesting that MRI-derived waveforms can be consistent in RR estimates across methods [15].

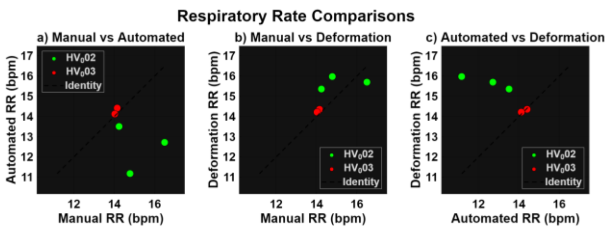

Fig 5 shows the RR scatterplots for each pairwise comparison – a) Manual-Automated, b) Manual-Deformation, c) Automated-Deformation

Abnormal Breathing

The three methods showed little agreement on abnormal intervals, as shown below: the manual method flagged 19, the automated method flagged 36, and the deformation method flagged 14. Only 5 intervals were common to all three, in contrast with [34], who reported 80% accuracy for deep learning-based detection against manual labels, suggesting that performance depends on ground truth definition.

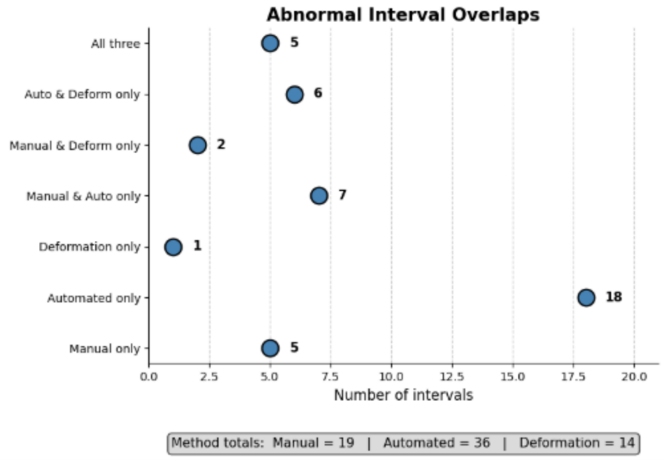

Fig 6 shows the abnormal interval overlaps.

Usability (DD):

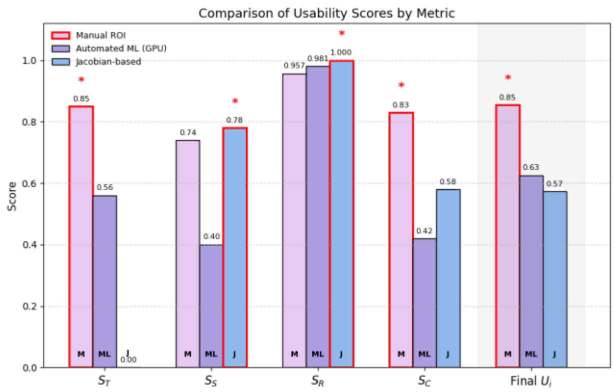

Fig 7. Comparison of usability component scores (S_T, S_S, S_R, S_C) and final usability score (U_i) across the three methods.

The quantitative comparison shown in Figure 6 indicated that the three methods achieved similar outputs but differed markedly in practical usability.

2.1 Time efficiency (S_T) and processing burden

To improve usability and reproducibility, CC introduced batch processing and manual ROI reuse; DD added timing functions for consistent runtime recording.

Manual ROI achieved the highest time-efficiency score ($S_T = 0.85$), with an average runtime of 36.83s per slice. This reflected its lightweight MATLAB workflow. In contrast, the Jacobian-based method scored ($S_T = 0.00$) (289.43 s per slice). This suggests that the main limitation of the registration-based approach was computational demand rather than consistency, as also reported in large-scale tissue expansion studies [35]. The ML method's 106.15s runtime revealed that data handling (52.89 s import stage) can double the total overhead, reducing efficiency.

2.2 Setup burden (S_S) and scalability

The ML method had the lowest setup score ($S_S = 0.40$) because of the additional effort required to configure the segmentation environment. However, it required less repeated manual input and was more scalable for larger datasets, consistent with the advantages of nnU-Net-based workflows, in which using segmentation masks removes the need for manual ROI drawing [36].

By comparison, Manual ROI ($S_S = 0.74$) and the Jacobian-based method ($S_S = 0.78$) both avoided the additional setup required for ML. This made both methods easier to implement in the current workflow than the ML approach.

2.3 Reproducibility (S_R) and practical reliability

All three methods showed high reproducibility ($S_R > 0.95$), with the Jacobian-based method achieving the highest score ($S_R = 1.000$). This was expected, as the determinant-based output was operator-independent. Manual ROI also showed strong reproducibility ($S_R = 0.957$), indicating that ROI-based respiratory surrogates can remain reliable despite their simpler implementation [37], although some operator dependence is still expected because region selection is subjective. The waveform comparisons shown below further support these reproducibility results. For the ML method, only minor fluctuations were observed between repeated inference runs.

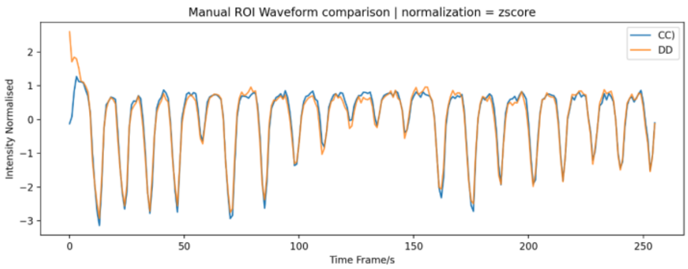

Fig 8. Inter-operator comparison of repeated waveform extraction for the Manual ROI method

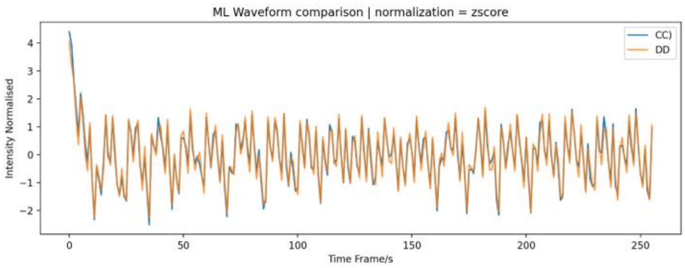

Fig 9. Inter-operator comparison of repeated waveform extraction for the Automated ML method.

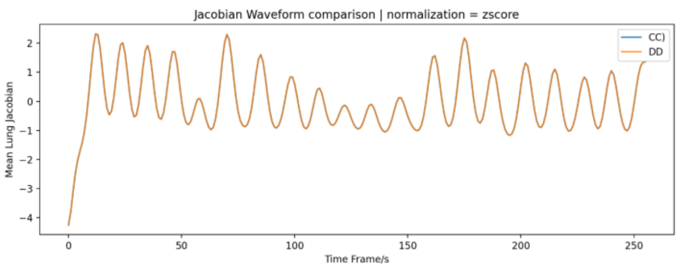

Fig 10. Inter-operator comparison of repeated waveform extraction for the Jacobian-based method.

Thus, method selection could not be based on reproducibility alone, but on the balance between runtime, setup burden, reproducibility, and computing demand. Therefore, Manual ROI achieved the highest final usability score.

Robustness (BB):

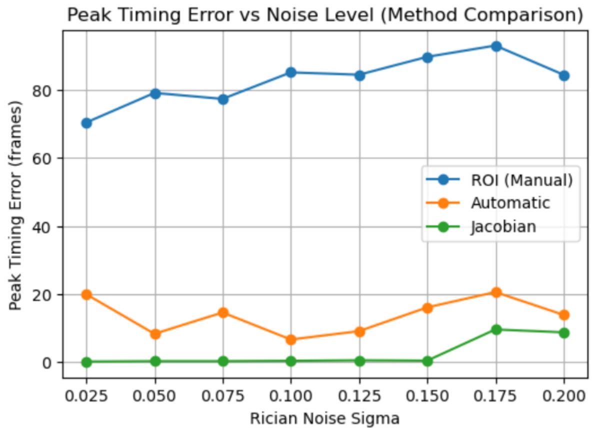

Fig 11. Peak timing error between the Rician-noisy and reference waveforms

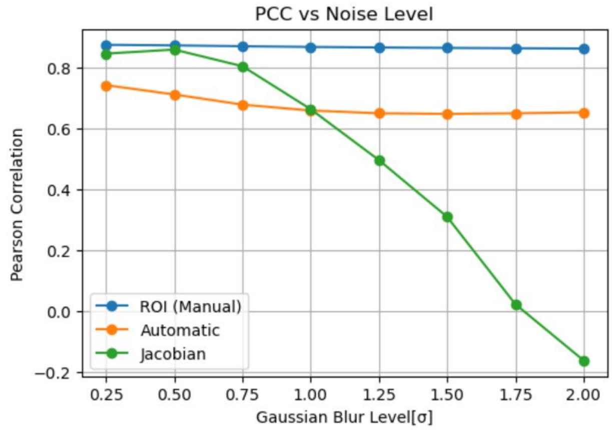

Fig 12. PCC between the Gaussian-noisy and reference waveform

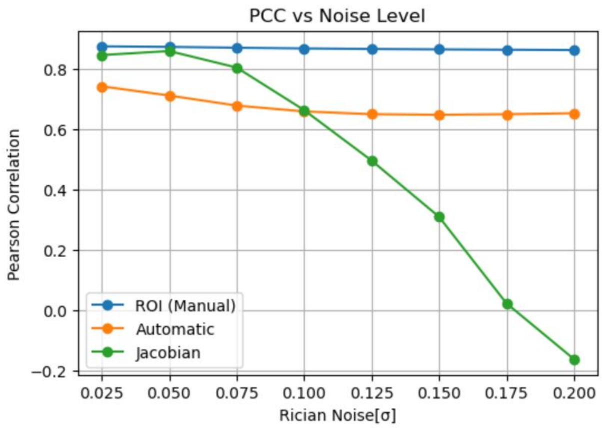

Fig 13. PCC between the Rician-noisy and reference waveforms

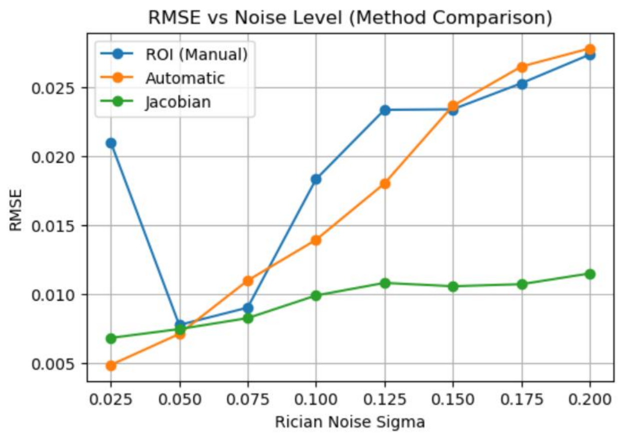

Fig 14. RMSE between the Rician-noisy and reference waveforms

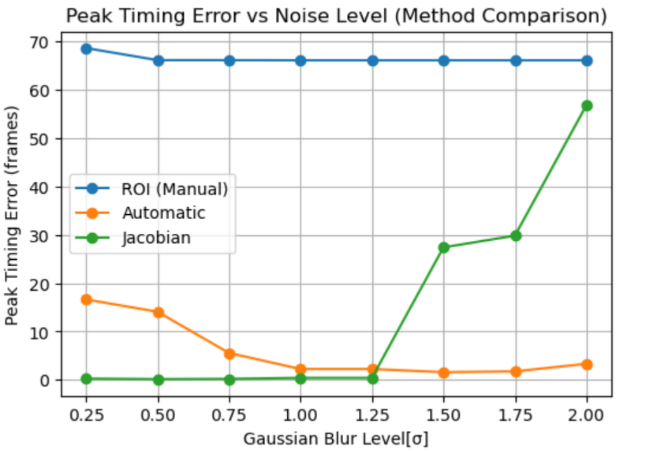

Fig 15. *Peak timing error between the Gaussian-noisy and reference waveforms*

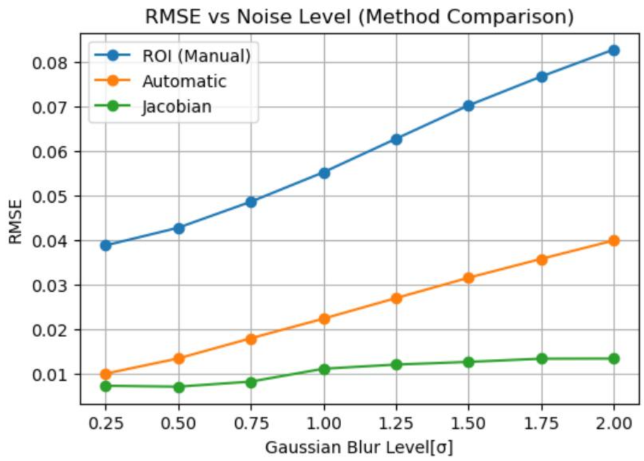

Fig 16. *RMSE between the Gaussian-noisy and reference waveforms*

The results demonstrate consistent trends across both Rician noise and Gaussian blur, revealing clear differences in robustness between the three methods. In terms of waveform shape, the ROI method maintains consistently high Pearson correlation across all levels of degradation (Fig. 12, Fig. 13), indicating strong robustness. However, this does not extend to temporal accuracy, as the ROI method exhibits consistently large peak timing errors (Fig. 11, Fig. 15), suggesting limited capability in capturing true respiratory dynamics [38][39].

The Jacobian method performs comparably under low degradation, achieving high correlation and minimal peak timing error (Fig. 11, Fig. 15), reflecting accurate deformation estimation and temporal alignment [40]. For both Rician noise and Gaussian blur, a clear threshold occurs—$\sigma \approx 0.1$ for noise and $\sigma \approx 1$ for blur (Fig. 12, Fig. 13)—beyond which correlation drops, peak timing error rises, and waveform inversion occurs, indicating registration failure [40].

The Automatic method shows more gradual degradation across metrics. Its correlation is lower than ROI, and peak timing error is higher than Jacobian under low-noise conditions, but it avoids abrupt failure (Fig. 12, Fig. 14), demonstrating stable performance across degradations.

RMSE analysis further highlights differences. ROI shows the largest RMSE despite high correlation (Fig. 14, Fig. 16), indicating sensitivity to intensity changes while preserving waveform shape. Jacobian maintains low RMSE even as correlation declines, producing signals of correct magnitude but incorrect structure, showing RMSE alone is insufficient [41].

Overall, ROI is robust but temporally limited, Jacobian is accurate under ideal conditions but sensitive to degradation, and Automatic provides balanced, stable performance across both degradation types.

IV. Conclusion

This project compared three methods for extracting respiratory waveforms from lung MRI: Manual ROI, Automatic Machine Learning, and the Jacobian method. All three produced useful waveforms, but each had distinct strengths and weaknesses, with no single method excelling in every category.

All methods captured overall respiratory rates, indicating they can represent normal breathing patterns reliably. However, agreement weakened for abnormal breathing intervals, suggesting they do not consistently detect unusual events.

Regarding usability, Manual ROI was quickest, easiest to implement, and highly reproducible. Machine Learning required more setup but scaled better for larger datasets. The Jacobian method provided detailed motion information and excellent reproducibility but had high computational cost and long runtimes.

Robustness showed trade-offs: ROI remained stable under noise and blur but lacked precise timing. Jacobian performed well with high-quality data but degraded with lower quality. Machine Learning offered balanced performance across varying conditions.

The study achieved its goal of a consistent comparison. Future work should include more patients, diseased lungs, stronger ground truth for abnormal breathing, and newer techniques like feature tracking, which follows visible structures such as the diaphragm to generate clinically interpretable waveforms, though it may not capture whole-lung motion.

In summary, method choice depends on the goal: Manual ROI for simplicity and speed, Machine Learning for scalable and balanced performance, and Jacobian for detailed motion analysis with high-quality images.

V. References

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VI. Appendices

Appendix A – Meeting Minutes

Group Tutors – Mina (Board member), Ben (group tutor)

Group Members – AA, BB, CC, DD and EE

A.1 Meeting 1: 3 February 2026

Attendees: All

Location/Time: Online, 13:10–13:50

Agenda & Discussion

- **Introduction** – Ben introduced himself (MSc UCL Hawkes Institute).

- **Code progress** (CC) – CC shared initial ML code set up and challenges.

- **Jacobian** (BB, AA) – Discussed Jacobian literature; noted potential for a less opaque methodology (less 'black box') than ML, with greater adaptability. However, the scope for the method in Lung MRI remains limited.

- **Dataset** – Ben confirmed that datasets had been uploaded to the ML folder.

- **Literature findings** (CC) – Noted that most papers focus on downstream applications (waveform → metrics, not MRI to waveform), but they provide insights into suitable comparison metrics.

- **Manual ROI** (CC, EE) – MATLAB ROI uploaded the previous day, no presentation yet.

- **Downstream applications** (EE) – Shared findings and discussed code.

Action Items

- **All**: Wrap up lit reviews; each sub-team to present methodology to the group on Tuesday.

- **Next supervisor meeting**: 24 February (agenda: present lit reviews and preliminary analysis; begin pipeline development).

Board Member Feedback

CC, all – Mina encouraged grounding all work in literature before presenting code.

A.2 Meeting 2: 24 February 2026

Attendees: All group members

Board Members: Module lead, group tutors

Location/Time: Teams (group only) 16:15–16:25; Zoom (with supervisors) 16:30–17:00

(**Mid-term Review**)

Agenda & Discussion

- **Pre-meeting (group only)** – Discussed presentation structure and set a post-reading week debrief for Thursday.

- **Mid-term review** (Zoom with supervisors and module lead) – All presented current progress in both individual goals and group goals. All mid-term review slides have been uploaded to the shared drive.

- **Feedback** – Module lead noted good collaboration and acknowledged that some groups were exceeding the 5-hour weekly effort.

Action Items

- **All**: Upload preliminary analysis (waveforms, code, etc.) to shared repository; complete stage 1 lit reviews.

- **Subgroups**: Synchronise to achieve personal goals outlined in slides.

Board Member Feedback

Module lead: commended teamwork; noted that the group appeared on track.

A.3 Meeting 3: 3 March 2026

Attendees: All (Mina present 14:00–14:30)

Location/Time: Teams, 14:00–15:00

Agenda & Discussion

- **Usability** (DD) – Presented qualitative findings; noted reproducibility issues and Jacobian boundary problems (previously raised by AA, BB).

- **Jacobian** (AA) – Presented Jacobian code overview; suggested using lung MRI GIFs for poster (using a projector).

- **Robustness** (BB) – Explained code limitations (tracks only y-axis, not true motion); noted waveforms are less noisy than ML.

- **Agreeability** (CC) – Presented draft pipeline for manual and ML methods (adaptable to Jacobian). Covered metrics, ROI selection, and overall progress.

Action Items

- **All**: Next week's objectives – AA/BB continue Jacobian/robustness (test noise); DD incorporate literature into usability; EE investigate manual ROI as per task (only CC has contributed to manual ROI so far, EE and CC were tasked on this together); CC continue agreeability pipeline and implement Ben's ROI feedback.

- **Next meetings**: group meeting Thursday/Friday.

Board Member Feedback

- **DD, all:** Mina advised that presentations should be visibly backed by citations.

- **CC**: Ben emphasised that manual ROI must be consistent across runs and confined to the region of interest, not the whole lung, to avoid vertical offset; shared examples.

- **BB, AA:** Ben suggested exploring open-source MATLAB registration packages.

A.4 Meeting 4: 17 March 2026

Attendees: All

Location/Time: Teams, 16:10–17:00

Agenda & Discussion

- **Coursework overview** – AA summarised deadlines.

- **Agreeability presentation** (CC) – CC presented the final pipeline.

- **Jacobian presentation** (AA) – AA presented the rationale and methodology of deformation-based extraction.

- **Robustness** (BB) – BB presented the choice of Rician noise and robustness methodology.

- **Usability** (DD) – DD presented an assessment of run times and ICC.

- **Fourth method** (EE) – EE gave an overview of the feature-tracking approach.

Action Items

- **All**: Finalise group report and poster for next week.

Board Member Feedback

- **CC**: Mina clarified the inverse correlation and discussed the Sun et al. reference.

- **BB**: Mina is keen on the idea and looking forward to seeing final results.

- **DD**: Mina confirmed whether a different ROI was used for manual-ROI ICC.

- **EE**: Mina and Ben requested a more detailed explanation of the anatomical structure tracked.

- **CC** (email): Mina encouraged more concise presentations to respect time; CC acknowledged and apologised for the overrun.

Appendix B – Code Summaries

Full code is available in the group repository – here makanakamazheke/LUNG-MRI-WAVEFORM-EXTRACTION: YEAR 3 PROJECT.

B.1 CC – Agreeability Pipeline

The following scripts were written and tested by CC as part of the project's quantitative comparison pipeline.

Script Name	Purpose
manual_waveform_extraction.m	Extracts respiratory waveforms using manual ROI (right hemi-diaphragm). Implements batch modes, ROI reuse, standardised saving (ROI, raw signal, normalisation data).
ML_waveform_extraction.m	Extracts waveforms using nnU-Net lung masks. Includes interactive and auto-accept modes; saves waveform and overlay images
comparison_all_methods.m	Loads manual, automated, deformation waveforms; computes per-slice and per-patient metrics (correlation, MAE, lag, phase agreement, cycle counts, RR, EE/EI differences); generates overlay plots, heatmaps, and summary tables
detect_abnormal_agreement.m	Runs RR-outlier detection on all three methods independently; computes pairwise overlaps; saves raw intervals for Gantt plots and CSV exports.
generate_robustness_waveforms.m	Processes noise and blur levels for a fixed slice (HV_002/raw_IM_0025) to generate CSV waveforms for BB's robustness analysis.

B.2 DD – Usability Comparison Pipeline

The following scripts were written and tested by DD as part of the project's quantitative usability comparison pipeline.

Script Name	Purpose
manualroiextraction_time.py	Extracts respiratory waveforms via interactive manual ROI selection; features integrated time-tracking to quantify manual labour duration as a key usability metric.
icc_usability_comparison_pipeline.py	Extracts respiratory waveforms from automated nnU-Net lung masks; records total processing time to evaluate algorithmic efficiency; saves waveform and overlay images
icc_usability_comparison_pipeline.py	Inter-operator validation : statistically validates intensity-based waveforms (`.csv`) against mean lung Jacobian sources (`.mat`) across the timeframe; computes agreement metrics including Pearson correlation (r), MAE, RMSE, and Intraclass Correlation Coefficient (ICC 2,1)
barchart.py	Generates standardised visualisations of usability scores (S_T, S_S, S_R, S_C); aggregates time efficiency and statistical accuracy to determine the final Usability Index (U_i) across the three methods.

B.3 AA – Jacobian Pipeline

The following scripts were written and tested by AA, as part of the project's quantitative usability comparison pipeline.

Script Name	Purpose
Deformation_Jacobian_Pipeline.ipynb	Creates its binary masks using nibabel and SimpleITK libraries. Uses a deformation field and the Jacobian determinant of this field to extract a respiratory waveform
Visualisation_Demons_Registration.ipynb	Images Lung MRI images visually through frame-by-frame so movement can be seen. Extracts a respiratory waveform using demons' registration which was used as a simplified Jacobian experiment.

B.4 AA + BB Deformation Pipeline incorporated with ML

Script Name	Purpose
Deformation_ML_Pipeline	To incorporate the ML masks into the Jacobian pipeline for accurate and consistent segmentation.

B.5 BB – Robustness Pipeline

The following scripts were written and tested by BB as part of the project's robustness pipeline.

Script Name	Purpose
Robust.ipynb	Takes 256 images(1 slice), and applies gaussian blur and Rician noise at 8 levels and saves these to path directories
Deformation_based_waveform_extraction_pipeline.ipynb	Takes a folder with 8 slices of varying noise and computes waveforms and saves them as csv files.
Metrics_calc.ipynb	Takes in csv files of waveforms of varying noise calculate and plots metrics(PCC, RMSE, PTE)

B.6 EE – Feature Tracking

The following scripts were written and tested by EE as part of further research.

Script Name	Purpose
feature_tracking_code.ipynb	Uses the diaphragm wall for feature tracking